Class 5: Data Vis with ggplot

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Background

There are lot's of ways to make plots in R. These include so-called "base R" (like the `plot()`) and add on packages like **ggplot2**.

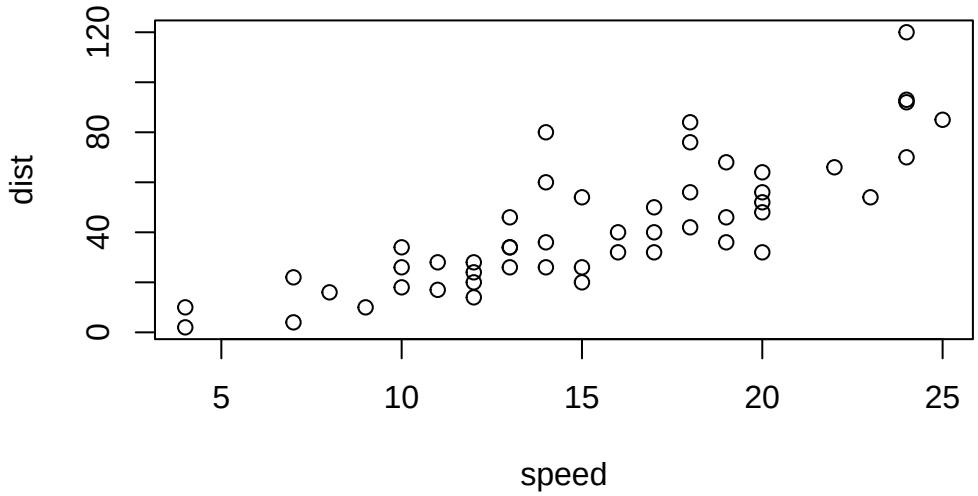
Let's make the same plot with these two graphics systems. We can use the inbuilt `cars` dataset:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

With "base R" we can simply:

```
plot(cars)
```



Now let's try ggplot. First I need to install the package using `install.packages("ggplot2")`.

> **N.B.** We never run an `install.packages()` in a code chunk otherwise we will re-install needlessly every time we render our document.

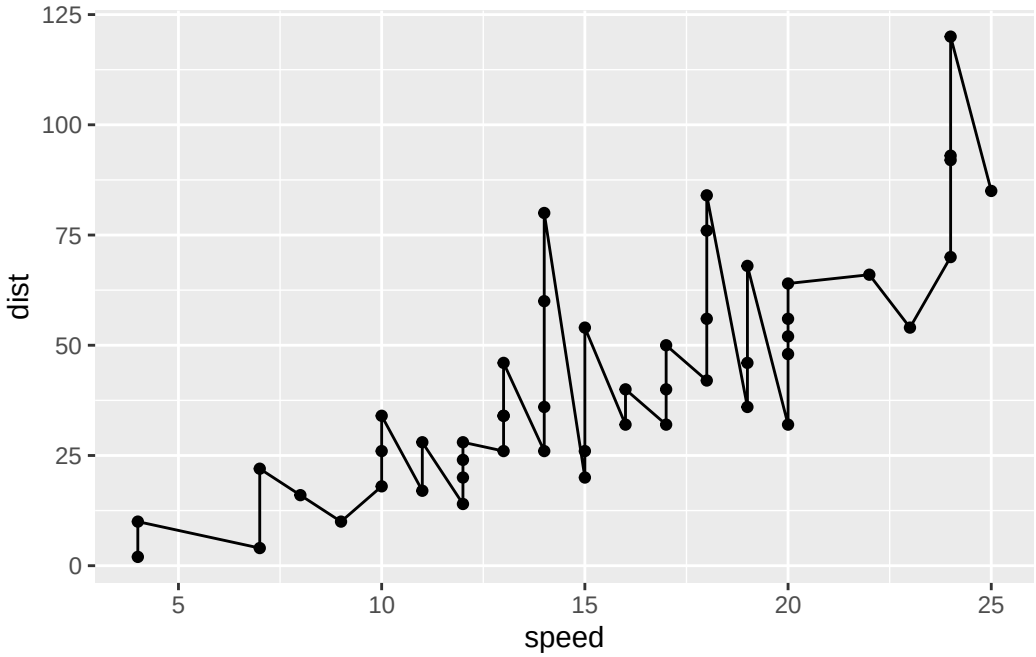
Everytime we want to use an add-on package we need to load it up with a call to `library()`

```
library(ggplot2)
ggplot(cars)
```

Every ggplot needs at least three things:

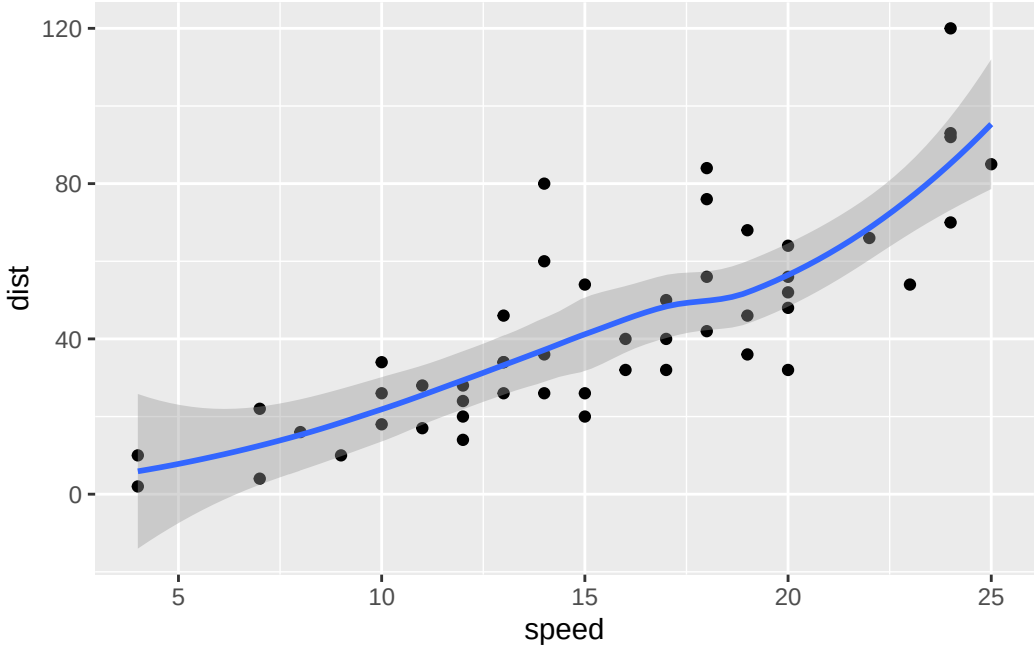
1. The **data** i.e. stuff to plot as a data.frame
2. The **aes** or aesthetics that map the data to the plot
3. The **geom_** or geometry i.e. the plot type such as points, lines, etc.

```
ggplot(cars) + aes(x=speed, y=dist) + geom_point() + geom_line()
```



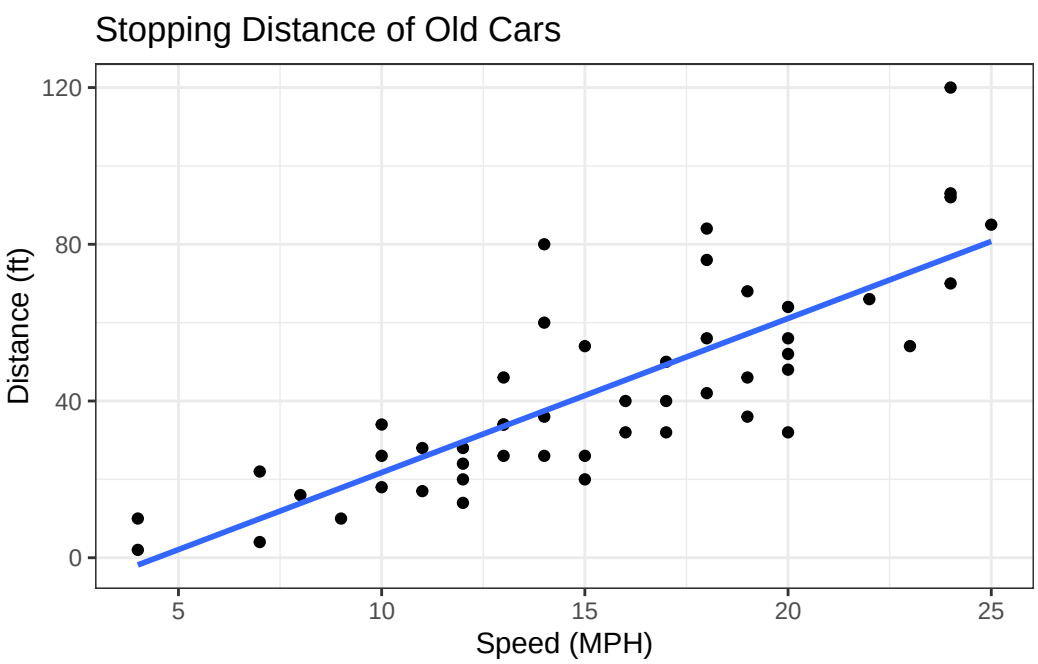

```
ggplot(cars) + aes(x=speed, y=dist) + geom_point() + geom_smooth()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



```r
ggplot(cars) + aes(x=speed, y=dist) + geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(x="Speed (MPH)",
       y = "Distance (ft)",
       title = "Stopping Distance of Old Cars") + theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Gene Expression Plot

Read some data on the effects of GLP-1 inhibitor (drug) on gene expression values.

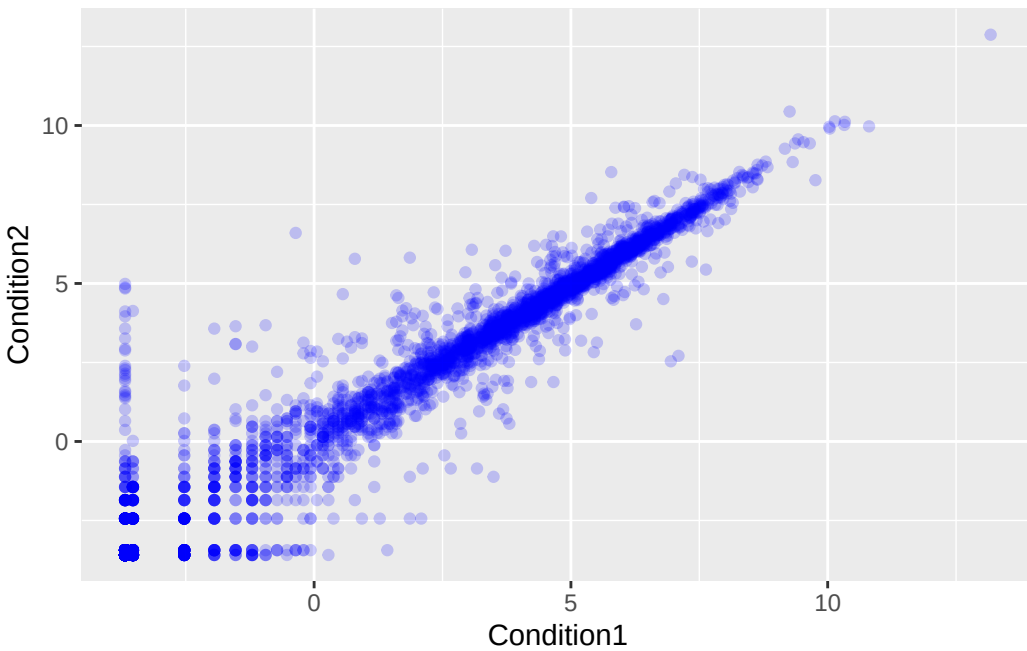
```r
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
```

```
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

Version 1 Plot - start simple by getting some ink on the page.

```
ggplot(genes) + aes(Condition1, Condition2) + geom_point(col="blue", alpha=0.2)
```

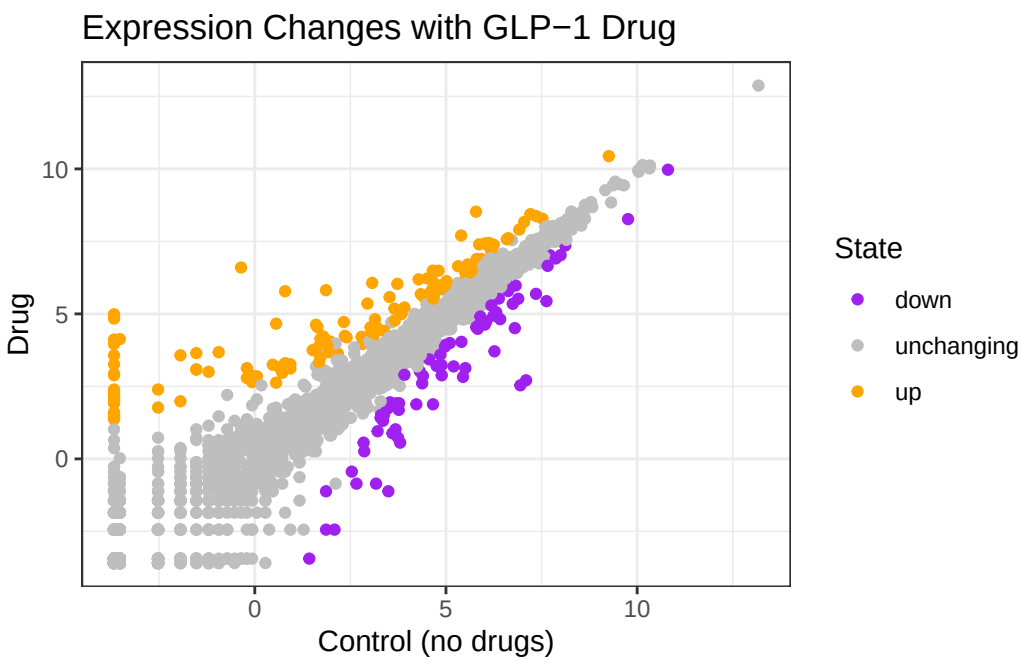


Let's color by `State` up, down or no change.

```
table(genes$State)
```

```
      down unchanging         up
        72       4997        127
```

```
ggplot(genes) + aes(Condition1, Condition2, col=State) + geom_point() +
  scale_color_manual(values = c("purple", "gray", "orange")) +
  labs(x="Control (no drugs)",
       y= "Drug",
       title = "Expression Changes with GLP-1 Drug") + theme_bw()
```



Going further with gapmider

Here we explore the famous `gapminder` dataset with some custom plots.

```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv
gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Q. How many rows does this dataset have?

```
nrow(gapminder)
```

```
[1] 1704
```

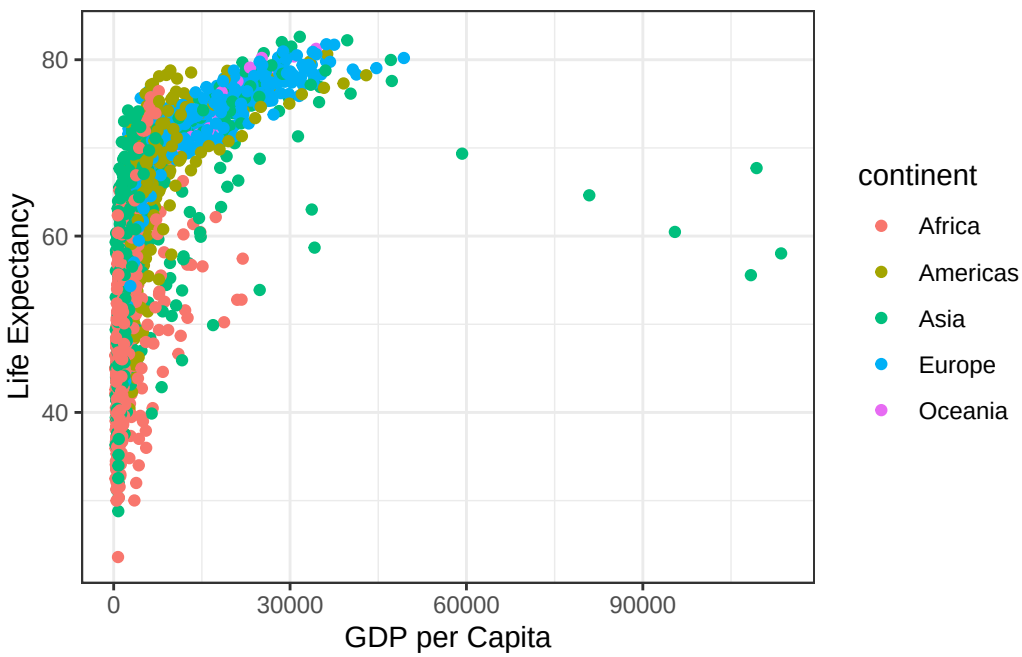
Q. How many different continents are in this dataset ?

```
table(gapminder$continent)
```

```
  Africa Americas     Asia   Europe  Oceania 
     624      300      396      360       24 
```

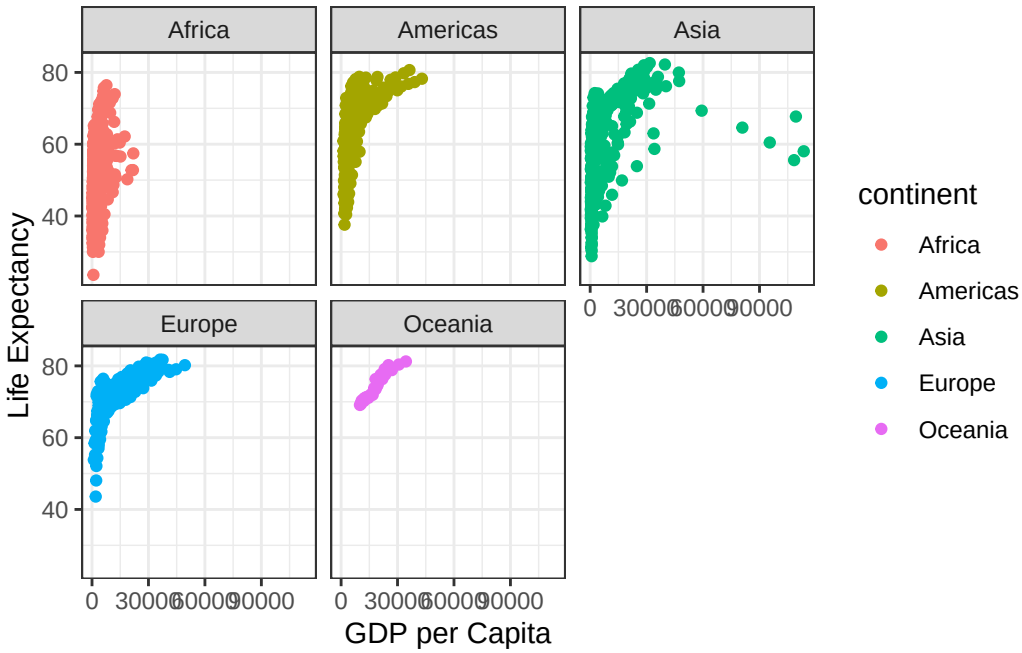
Version 1 plot GDP vs LifeExp for all rows

```
ggplot(gapminder) + aes(gdpPercap, lifeExp, col=continent) + geom_point() +
  labs(x="GDP per Capita", y="Life Expectancy") + theme_bw()
```



I want to see a plot for each continent - in ggplot lingo this is called "Faceting"

```
ggplot(gapminder) + aes(gdpPercap, lifeExp, col=continent) + geom_point() +
  labs(x="GDP per Capita", y="Life Expectancy") +
  theme_bw() + facet_wrap(~continent)
```



First look at the dplyr package

Another add-on package with a function called `filter()` that we want to use.

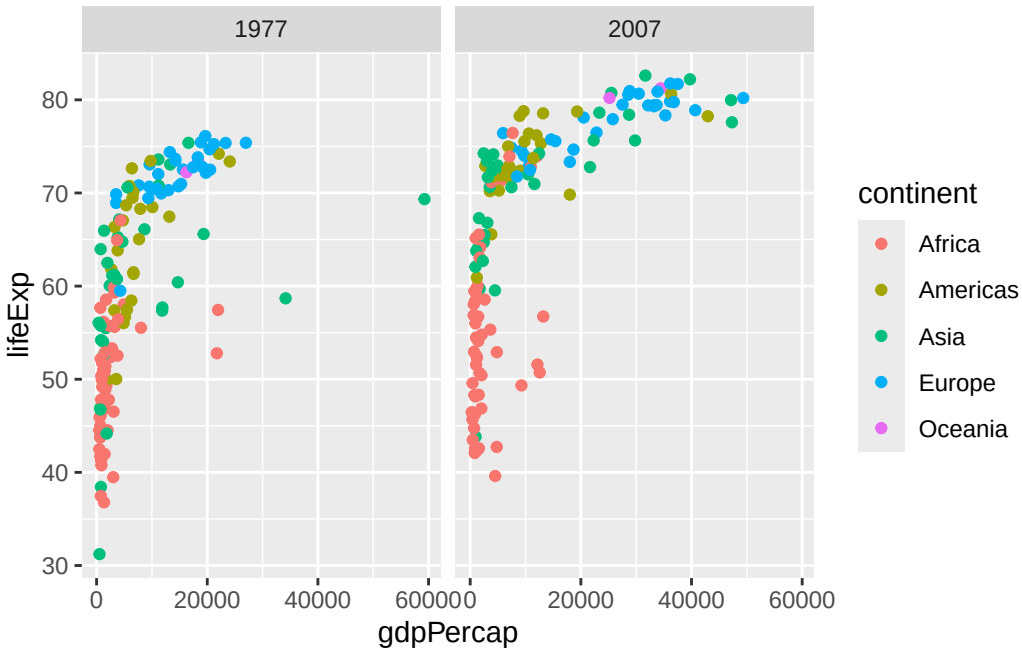
```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
filter(gapminder, year == 2007, country == "Ireland")
```

```
  country continent year lifeExp     pop gdpPercap
1 Ireland    Europe 2007  78.885 4109086     40676
```

```r
filter(gapminder, year == 2007, country == "United States")
```

```
        country continent year lifeExp       pop gdpPercap
1 United States  Americas 2007  78.242 301139947  42951.65
```

```r
input <- filter(gapminder, year == 2007 | year == 1977)

ggplot(input) + aes(gdpPercap, lifeExp, col=continent) +
  geom_point() + facet_wrap(~year)
```